

Evaluation Indicators for Speaker Identification Accuracy

Closed-Set, Open-Set, Verification, Calibration, Security, and Statistical Evaluation

arXiv-style technical report

Primary problem	Speaker identification (interpreting “phonologist identification” as speaker identification)
Core distinction	1: N identification is not the same as 1:1 speaker verification
Main closed-set metrics	Top-1, Top- k , CMC, macro recall/F1, balanced accuracy, confusion analysis
Main open-set metrics	DIR@FAR / TPIR@FPIR, unknown rejection, gallery-size sensitivity
Supporting embedding metrics	EER, DET/ROC, minDCF, calibration and C_{llr}
Operational extensions	latency, duration/SNR robustness, anti-spoof t-DCF, confidence intervals

August 2026

Technical evaluation methodology for deep-learning speaker-identification systems

Abstract

The phrase “speaker-identification accuracy” is often reduced to a single percentage, but no single metric adequately characterizes a modern speaker-identification system. Closed-set identification asks the system to select one identity from a known roster and is naturally measured with Top-1/Top- k accuracy and cumulative matching characteristics. Open-set identification is harder: the system must perform $1:N$ search and also reject speakers who are not enrolled; it therefore requires a joint detection-and-identification metric such as Detection and Identification Rate at a specified False Alarm Rate (DIR@FAR). Speaker-verification metrics such as equal error rate (EER), DET curves, and minimum detection cost function (minDCF) remain valuable because most modern identification systems use verification-style speaker embeddings, but these metrics do not by themselves measure $1:N$ identification. Calibration metrics such as C_{llr} measure whether output scores have probabilistic meaning, while security-oriented systems additionally require spoofing-aware measures such as tandem DCF (t-DCF). This report derives the principal metrics, explains their operating assumptions, identifies common misuse, and proposes minimum metric sets for research, access control, watchlists, dynamic enrollment, and speaker-identification systems based on deep neural embeddings. Particular emphasis is placed on gallery size, number of enrollment utterances, probe duration, class imbalance, confidence intervals, and threshold selection. The central recommendation is to define the evaluation protocol first and then report a complementary metric set; reporting only Top-1 accuracy or only EER is insufficient for most practical speaker-identification deployments.

Keywords: speaker identification, speaker recognition, open-set identification, Top- k , CMC, DIR@FAR, EER, minDCF, DET, calibration, C_{llr} , t-DCF, voice biometrics.

Contents

1	Scope and terminology	1
2	The evaluation protocol comes before the metric	1
3	Closed-set speaker-identification metrics	2
3.1	Top-1 accuracy	2
3.2	Top- k accuracy	2
3.3	Cumulative Matching Characteristic (CMC)	3
3.4	Mean rank and mean reciprocal rank	3
3.5	Mean Average Precision (mAP)	4
4	Class-wise accuracy, precision, recall, and F1	4
4.1	Confusion matrix	4
5	Open-set speaker identification: the critical operational metrics	5
5.1	Problem definition	5
5.2	Detection and Identification Rate (DIR)	5
5.3	False Alarm Rate for unknown probes	6
5.4	DIR@FAR	6
5.5	TPIR/FPIR/FNIR terminology	6

5.6	Open Set Classification Rate (OSCR)	7
6	Gallery size, enrollment count, and the false-alarm problem	7
7	Speaker-verification metrics as embedding diagnostics	7
7.1	False alarm and miss rates	7
7.2	Equal Error Rate (EER)	8
7.3	ROC and DET curves	8
7.4	ROC AUC	8
8	Detection Cost Function and minDCF	9
8.1	Generic detection cost	9
8.2	NIST SRE21 perspective	9
9	Calibration metrics	9
9.1	C_{llr}	9
9.2	Actual DCF versus minDCF	9
9.3	Brier score and negative log-likelihood	10
9.4	Expected Calibration Error (ECE)	10
10	Security-aware evaluation and t-DCF	10
11	Embedding-space diagnostic indicators	10
11.1	Within- and between-speaker dispersion	10
11.2	d' separation	11
11.3	Silhouette score	11
12	Worked metric examples	11
12.1	Closed-set example	11
12.2	Open-set example	11
12.3	Verification-cost example	12
13	Statistical uncertainty and significance	12
13.1	Confidence interval for accuracy	12
13.2	Paired system comparison	12
13.3	The NIST “Rule of 30” heuristic	12
14	Condition-stratified performance indicators	13
14.1	Probe duration	13
14.2	SNR and acoustic environment	13

14.3 Channel/device	13
14.4 Language and cross-lingual condition	13
14.5 Enrollment count	13
14.6 Gallery size	13
15 Metric pitfalls and common reporting errors	13
16 Recommended metric sets by application	14
17 A recommended evaluation report template	15
17.1 Protocol table	15
17.2 Primary accuracy table	15
17.3 Embedding diagnostic table	15
17.4 Robustness table	15
17.5 Uncertainty	16
18 Metric selection for deep-learning model development	16
19 Interpretation of benchmark results	16
19.1 VoxCeleb closed-set identification	16
19.2 VoxBlink2 open-set identification	16
19.3 VoxWatch	16
20 Final recommendations	17
21 Conclusion	17
A Compact formula reference	18
B Checklist for an evaluation section in a paper	18

List of Tables

1	Common metric mistakes and corrective practice.	14
2	Recommended minimum evaluation indicators.	14

List of Figures

1	Recommended evaluation sequence. Metrics are selected after the operational task and protocol are fixed.	2
2	Illustrative CMC curves. System A has better rank-1 performance and reaches high cumulative identification rate with fewer candidates. Curves are schematic.	3
3	Illustrative two-class confusion matrix. In a real speaker-identification system the matrix is $C \times C$ and often strongly sparse.	5
4	VoxBlink2-reported DIR at FAR=0.001 versus gallery size, for 1/3/5 enrollment utterances. The large-gallery drop shows why gallery size and enrollment count must accompany any open-set accuracy claim [2].	6
5	Illustrative threshold trade-off. Lower miss at the same false-alarm rate is better. A formal DET curve uses normal-deviate axis transforms; the figure is a simplified log-FAR visualization.	8

1 Scope and terminology

This report interprets the requested term *phonologist identification* as *speaker identification*, consistent with the surrounding technical context. Speaker identification answers “Which enrolled speaker produced this utterance?” Modern systems usually map an utterance to a speaker embedding and then compare that embedding with an enrollment gallery.

The most important evaluation mistake is to mix three different tasks:

Closed-set identification

Every probe is guaranteed to belong to one of the enrolled speakers. The system must rank or choose the correct speaker.

Open-set identification

Some probes are from enrolled speakers and some are unknown. The system must identify known speakers and reject unknown speakers.

Speaker verification

A 1:1 binary trial asks whether a probe and a claimed/enrolled speaker are the same person.

VoxCeleb explicitly used Top-1 and Top-5 accuracy for identification [1], whereas modern open-set speaker-identification benchmarks such as VoxBlink2 use DIR and FAR because a simple closed-set accuracy cannot express unknown-speaker rejection [2]. NIST speaker-recognition evaluations, which are primarily verification/detection evaluations, emphasize detection cost, calibration, and DET curves rather than 1:N rank accuracy [5, 7].

2 The evaluation protocol comes before the metric

A metric is meaningful only together with its protocol. At minimum, a speaker-identification result should state:

- number of enrolled speakers N_G ;
- number of enrollment utterances per speaker M_E ;
- whether the task is closed-set or open-set;
- number of known probes $|K|$ and unknown probes $|U|$;
- probe duration distribution;
- whether speakers, sessions, recordings, or source videos overlap across train/dev/test;
- score type (cosine, PLDA, classifier logit, calibrated log-likelihood ratio);
- how multiple enrollment utterances are combined;
- how any threshold is chosen and which development set is used;
- whether score normalization, fusion, or calibration is applied;
- whether evaluation is stratified by device, SNR, language, duration, sex/gender, or other relevant factors.

These variables can change the reported number more than an architecture change. VoxWatch specifically documents a “false-alarm problem” in open-set speaker identification as watchlist size grows: the maximum impostor score tends to rise because more enrolled candidates are searched [4]. VoxBlink2 similarly evaluates small, medium, and large galleries and demonstrates substantial degradation at large gallery size [2].

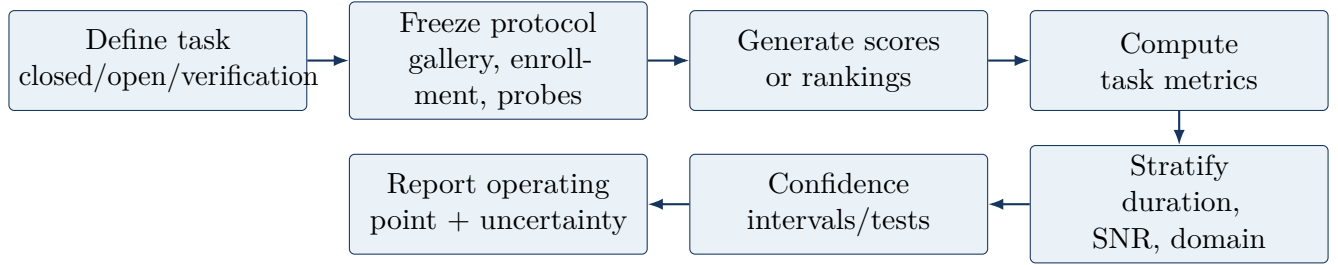


Figure 1: Recommended evaluation sequence. Metrics are selected after the operational task and protocol are fixed.

3 Closed-set speaker-identification metrics

3.1 Top-1 accuracy

Let N be the number of probe utterances and let y_i be the true speaker. If $\hat{y}_i$ is the highest-scoring gallery identity, Top-1 accuracy is

$$\text{Acc}_1 = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i). \quad (1)$$

Top-1 error rate is

$$\text{Err}_1 = 1 - \text{Acc}_1. \quad (2)$$

This metric is intuitive and appropriate when the system must always return exactly one enrolled speaker. The original VoxCeleb identification experiment reported Top-1 and Top-5 accuracy, demonstrating the standard use of rank-based identification metrics [1].

3.2 Top- k accuracy

Let $\mathcal{R}_i(k)$ be the set of the k highest-ranked identities for probe i . Then

$$\text{Acc}_k = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i \in \mathcal{R}_i(k)). \quad (3)$$

Top- k is useful when a downstream analyst can inspect a shortlist, or when identification is a retrieval stage before more expensive re-scoring. It should not substitute for Top-1 when automatic final identity selection is required.

3.3 Cumulative Matching Characteristic (CMC)

Let r_i be the rank of the true speaker for probe i , where rank 1 is best. The CMC identification rate at rank k is

$$\text{CMC}(k) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(r_i \leq k). \quad (4)$$

Thus

$$\text{CMC}(1) = \text{Top1}. \quad (5)$$

CMC is more informative than one Top- k number because it shows the entire retrieval curve. It has also been used recently by NIST for measuring speaker-identity leakage in de-identification systems, where a high CMC hit rate indicates that original identity remains recoverable from a candidate list [11].

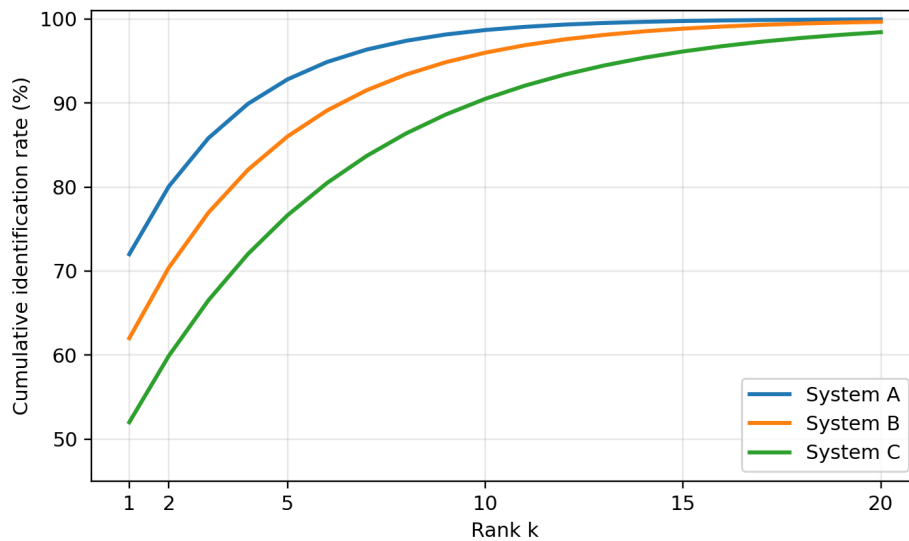


Figure 2: Illustrative CMC curves. System A has better rank-1 performance and reaches high cumulative identification rate with fewer candidates. Curves are schematic.

3.4 Mean rank and mean reciprocal rank

Mean rank is

$$\bar{r} = \frac{1}{N} \sum_i r_i, \quad (6)$$

while mean reciprocal rank is

$$\text{MRR} = \frac{1}{N} \sum_i \frac{1}{r_i}. \quad (7)$$

MRR rewards moving the correct speaker toward the top of the list and is useful in retrieval-oriented systems. It is less standard than CMC in speaker identification and should be reported as supplementary, not as the only primary metric.

3.5 Mean Average Precision (mAP)

mAP becomes meaningful when a query may have multiple relevant gallery items (for example, multiple enrollment clips retained separately rather than averaged into one prototype). For query q ,

$$\text{AP}_q = \frac{1}{R_q} \sum_{k=1}^{N_G} P_q(k) \text{rel}_q(k), \quad (8)$$

where R_q is the number of relevant gallery items, $P_q(k)$ is precision at cutoff k , and $\text{rel}_q(k)$ indicates whether the item at rank k belongs to the correct speaker. Then

$$\text{mAP} = \frac{1}{Q} \sum_q \text{AP}_q. \quad (9)$$

When the gallery has exactly one prototype per speaker, CMC and Top- k are usually more natural.

4 Class-wise accuracy, precision, recall, and F1

A single global accuracy can hide a subset of speakers that performs poorly. For each speaker c , define one-vs-rest counts TP_c, FP_c, FN_c, TN_c .

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad (10)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad (11)$$

$$F1_c = \frac{2 \text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (12)$$

Macro averages give each speaker equal weight:

$$F1_{macro} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad R_{macro} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c. \quad (13)$$

For single-label multiclass classification, micro-averaged precision/recall/F1 collapse to the same value as accuracy, so macro metrics add more information when class counts are unequal.

Balanced accuracy is

$$\text{BAcc} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c. \quad (14)$$

It is therefore the macro average of per-speaker recall and is useful when some speakers have many more probes than others.

4.1 Confusion matrix

A $C \times C$ matrix M contains

$$M_{ij} = \#\{\text{probes from speaker } i \text{ predicted as } j\}. \quad (15)$$

The confusion matrix is not a scalar metric, but it is essential for diagnosing systematic identity pairs. In large galleries, list the worst off-diagonal confusions or cluster the matrix rather than printing thousands of columns.

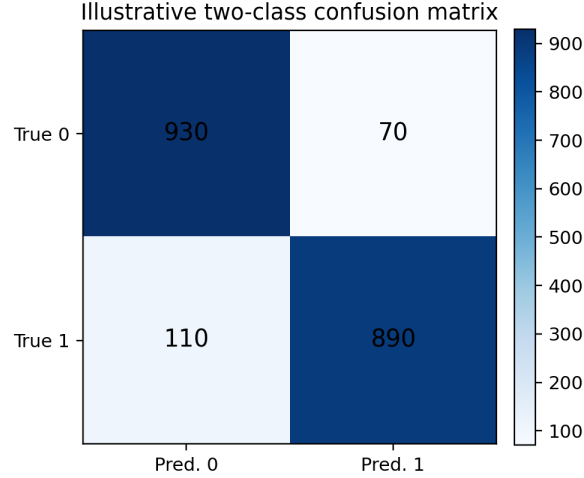


Figure 3: Illustrative two-class confusion matrix. In a real speaker-identification system the matrix is $C \times C$ and often strongly sparse.

5 Open-set speaker identification: the critical operational metrics

5.1 Problem definition

Let the gallery be G , the known probe set be K , and the unknown probe set be U . A probe $p \in K$ has a true enrolled identity g_p . An unknown probe $u \in U$ has no valid gallery identity.

For score $s(p, g)$ define

$$s^*(p) = \max_{g \in G} s(p, g). \quad (16)$$

The system returns a gallery identity if $s^*(p) \geq \theta$ and returns “unknown” otherwise.

5.2 Detection and Identification Rate (DIR)

For Top- k open-set identification,

$$\text{DIR}(\theta, k) = \frac{|\{p \in K : r(p) \leq k \wedge s(p, g_p) \geq \theta\}|}{|K|}. \quad (17)$$

For the common Top-1 case,

$$\text{DIR}(\theta, 1) = \frac{|\{\text{known probes correctly identified and above threshold}\}|}{|K|}. \quad (18)$$

Important implementation note. The typeset VoxBlink2 paper prints a rank inequality with the opposite direction, but its text states that rank 1 is the best match and its released evaluation code computes DIR from *correct Top-1 classification plus threshold acceptance*; the operational top- k condition is therefore $r(p) \leq k$ [2, 3].

5.3 False Alarm Rate for unknown probes

VoxBlink2 defines open-set false alarm rate as

$$\text{FAR}(\theta) = \frac{|\{u \in U : s^*(u) \geq \theta\}|}{|U|}. \quad (19)$$

This is the proportion of unknown speakers incorrectly assigned to some enrolled identity.

5.4 DIR@FAR

Choose an operational FAR target α such as 10^{-3} or 10^{-2} and report the maximum or interpolated DIR at a threshold that satisfies the FAR constraint:

$$\text{DIR@FAR} = \alpha. \quad (20)$$

This is often the best single-number metric for practical open-set SID because it combines correct identity ranking and unknown rejection. VoxBlink2 uses this measure in galleries ranging from 60 to 6,000 speakers and demonstrates that the same model can have very different operational accuracy as gallery size changes [2].

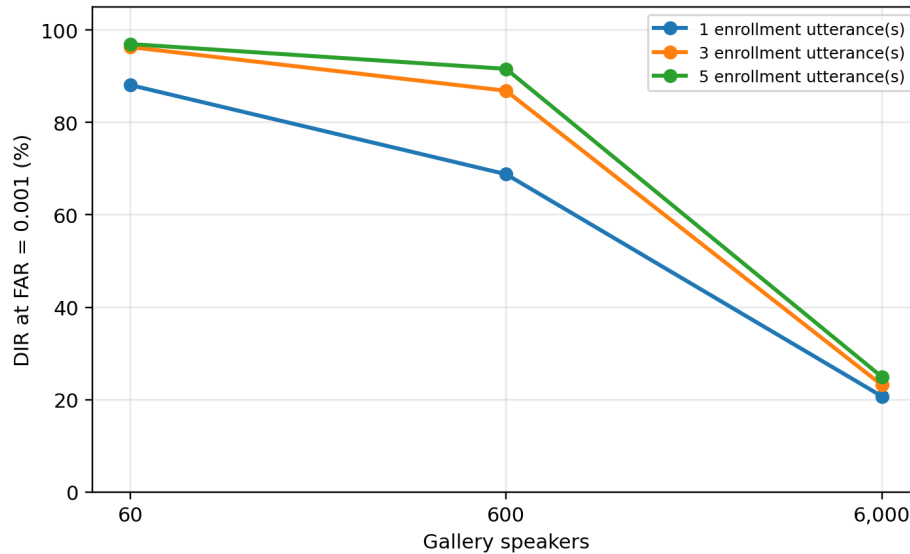


Figure 4: VoxBlink2-reported DIR at FAR=0.001 versus gallery size, for 1/3/5 enrollment utterances. The large-gallery drop shows why gallery size and enrollment count must accompany any open-set accuracy claim [2].

5.5 TPIR/FPIR/FNIR terminology

General biometric identification literature often uses:

- **False Positive Identification Rate (FPIR)**: fraction of unknown probes assigned to an enrolled identity;
- **True Positive Identification Rate (TPIR)**: fraction of known probes correctly identified above threshold, normally at a specified rank;

- **False Negative Identification Rate (FNIR)**: $1 - \text{TPIR}$ for the specified protocol.

For Top-1 open-set SID under matching definitions,

$$\text{TPIR} \approx \text{DIR}, \quad \text{FPIR} \approx \text{FAR}. \quad (21)$$

Terminology should be defined explicitly because speaker papers often use DIR/FAR while other biometric communities use TPIR/FPIR.

5.6 Open Set Classification Rate (OSCR)

OSCR-style plots show a correct classification rate for known samples against a false positive rate for unknown samples as the rejection threshold is swept. The concept is close to DIR-versus-FAR. For speaker identification, DIR@FAR is usually easier to interpret operationally because both the rank criterion and rejection threshold are explicit.

6 Gallery size, enrollment count, and the false-alarm problem

Open-set evaluation must vary at least three axes:

1. **Gallery size** N_G . Larger galleries create more opportunities for an unknown probe to obtain a spuriously high maximum similarity. VoxWatch shows this watchlist false-alarm effect and studies calibration/fusion as mitigation [4].
2. **Enrollment count** M_E . Averaging multiple enrollment embeddings reduces session variance. VoxBlink2 reports substantial improvement as enrollment utterances increase from 1 to 3 to 5 [2].
3. **Probe duration**. Short probes provide less phonetic coverage and noisier speaker statistics; thresholds and error rates may change with duration.

A publication should not state simply “DIR@FAR=90%” without also stating the gallery size, enrollment count, FAR operating point, and duration distribution.

7 Speaker-verification metrics as embedding diagnostics

Most deep speaker-identification systems are built from embeddings also used for 1:1 verification. Verification metrics are therefore valuable diagnostics of representation quality, but they do not replace identification metrics.

7.1 False alarm and miss rates

For verification threshold θ :

$$P_{fa}(\theta) = P(s \geq \theta \mid \text{non-target}), \quad (22)$$

$$P_{miss}(\theta) = P(s < \theta \mid \text{target}). \quad (23)$$

False rejection rate (FRR) is often used synonymously with P_{miss} ; false acceptance rate (FAR) is often used for P_{fa} . Because “FAR” is also used in open-set identification, reports should define it unambiguously.

7.2 Equal Error Rate (EER)

EER is the operating point where

$$P_{fa}(\theta_{eer}) = P_{miss}(\theta_{eer}). \quad (24)$$

The common error value is EER. NIST describes EER as intuitive but has historically emphasized cost-weighted operating points because many applications do not operate where the two errors are equally costly [6]. BOSARIS provides a principled treatment for computing EER and large-scale detection-cost metrics [8].

EER is useful for comparing embedding discrimination, but it is:

- a 1:1 metric;
- insensitive to score calibration because the threshold is optimized;
- not tied to the application’s target prior or error costs;
- not a measure of large-gallery 1: N false identification.

7.3 ROC and DET curves

A ROC curve sweeps threshold and plots detection rate against false-alarm rate. Speaker-recognition evaluations often use a Detection Error Tradeoff (DET) curve, plotting miss and false-alarm rates on normal-deviate axes. NIST introduced DET curves to make operating-region differences easier to inspect [5, 6].

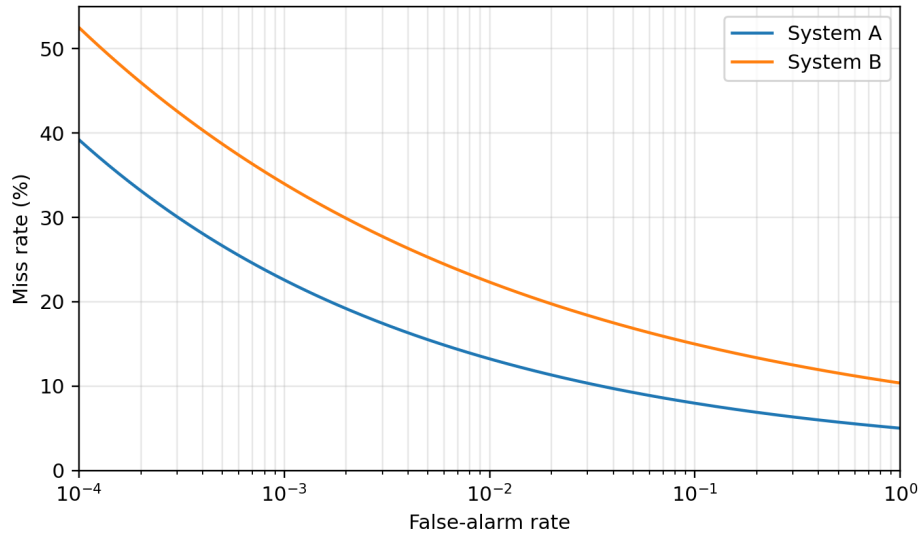


Figure 5: Illustrative threshold trade-off. Lower miss at the same false-alarm rate is better. A formal DET curve uses normal-deviate axis transforms; the figure is a simplified log-FAR visualization.

7.4 ROC AUC

Area under the ROC curve summarizes discrimination over all thresholds. It is useful for broad comparisons, but a system with excellent global AUC can still be poor at the very low false-alarm region required by a security application. Always report the relevant low-FAR operating point in addition to AUC.

8 Detection Cost Function and minDCF

8.1 Generic detection cost

For target prior P_{tar} , miss cost C_{miss} , and false-alarm cost C_{fa} ,

$$C_{det}(\theta) = C_{miss} P_{miss}(\theta) P_{tar} + C_{fa} P_{fa}(\theta) (1 - P_{tar}). \quad (25)$$

A common normalized version is

$$C_{norm}(\theta) = \frac{C_{det}(\theta)}{\min(C_{miss} P_{tar}, C_{fa} (1 - P_{tar}))}. \quad (26)$$

Then

$$\text{minDCF} = \min_{\theta} C_{norm}(\theta). \quad (27)$$

minDCF measures the best discrimination achievable for a specified application cost/prior model. Because the threshold is optimized on the evaluation score distribution, minDCF does not by itself measure calibration.

8.2 NIST SRE21 perspective

NIST SRE21 uses an official primary cost built from normalized detection costs at specified operating points and averages the result across defined partitions; it also reports a minimum detection cost for comparison [7]. The important design principle is that error costs and target priors belong in the metric when the deployment has asymmetric consequences.

9 Calibration metrics

A speaker system can rank trials well but produce badly calibrated scores. If scores are used to make risk-based decisions, fuse systems, or transfer a threshold between datasets, calibration matters.

9.1 C_{llr}

For calibrated log-likelihood-ratio scores ℓ , a common speaker-detection calibration measure is

$$C_{llr} = \frac{1}{2 \ln 2} \left[\frac{1}{N_{tar}} \sum_{i \in tar} \ln(1 + e^{-\ell_i}) + \frac{1}{N_{non}} \sum_{j \in non} \ln(1 + e^{\ell_j}) \right]. \quad (28)$$

Lower is better. C_{llr} is information-theoretic and can be decomposed conceptually into discrimination and calibration contributions; it is designed for systems whose outputs have likelihood-ratio interpretation [9].

9.2 Actual DCF versus minDCF

If a score is a calibrated log-likelihood ratio, the Bayes decision threshold follows directly from costs and priors. The *actual* DCF at that threshold reflects both discrimination and calibration, whereas minDCF optimizes the threshold and mainly reflects discrimination. Reporting both is therefore informative.

9.3 Brier score and negative log-likelihood

For applications that output posterior probabilities p_i , supplementary proper-scoring metrics include

$$\text{Brier} = \frac{1}{N} \sum_i (p_i - y_i)^2 \quad (29)$$

and binary negative log-likelihood. They are not the traditional headline speaker-recognition metrics, but they are useful when a downstream system consumes probabilities rather than rankings.

9.4 Expected Calibration Error (ECE)

ECE bins probability outputs and compares empirical accuracy with average confidence in each bin. It is popular in general deep learning but is sensitive to bin choice and does not replace likelihood-ratio calibration metrics in speaker detection. Use it only as supplementary evidence.

10 Security-aware evaluation and t-DCF

Speaker identification used for access control can be attacked with replay, voice conversion, text-to-speech, and deepfake audio. Identity accuracy on bona-fide speech is therefore insufficient.

The tandem detection cost function evaluates a spoofing countermeasure (CM) used together with an automatic speaker verifier (ASV). For a fixed ASV system, the normalized t-DCF can be expressed in the form

$$\text{tDCF}(\theta_{CM}) = C_0 + C_1 P_{miss}^{CM}(\theta_{CM}) + C_2 P_{fa}^{CM}(\theta_{CM}), \quad (30)$$

where coefficients depend on ASV error rates, priors, and costs. The original t-DCF work argues that countermeasure EER alone may rank systems differently from the risk of the combined ASV+CM system [10].

For a secure SID deployment, report at least:

- bona-fide identification DIR@FAR or Top-1;
- spoof-countermeasure EER or attack-class metrics;
- t-DCF (or an equivalent end-to-end security cost) if the CM gates identity acceptance;
- attack-specific results for replay, TTS, and voice conversion where relevant.

11 Embedding-space diagnostic indicators

Embedding geometry measures can help explain failures, but they are not substitutes for task-level accuracy.

11.1 Within- and between-speaker dispersion

Let $\mathbf{z}_{ci}$ be embedding i from speaker c and $\boldsymbol{\mu}_c$ its speaker centroid. Within-speaker scatter is

$$S_W = \frac{1}{N} \sum_c \sum_i \|\mathbf{z}_{ci} - \boldsymbol{\mu}_c\|_2^2. \quad (31)$$

Between-speaker scatter can be summarized as

$$S_B = \frac{1}{C} \sum_c \|\boldsymbol{\mu}_c - \boldsymbol{\mu}\|_2^2. \quad (32)$$

A diagnostic Fisher ratio is

$$J = \frac{S_B}{S_W}. \quad (33)$$

Higher separation is generally desirable, but the metric depends on embedding scale unless embeddings are normalized.

11.2 d' separation

For target and non-target score distributions with means μ_t, μ_n and variances σ_t^2, σ_n^2 ,

$$d' = \frac{|\mu_t - \mu_n|}{\sqrt{\frac{1}{2}(\sigma_t^2 + \sigma_n^2)}}. \quad (34)$$

Higher d' indicates stronger score separation. It is easy to interpret but assumes that a mean/variance summary is informative; heavy tails are especially important in very large galleries.

11.3 Silhouette score

Silhouette analysis can assess speaker clustering, but its Euclidean assumptions and sensitivity to sample composition mean it is best used for debugging. t-SNE/UMAP visual separation is not an accuracy metric.

12 Worked metric examples

12.1 Closed-set example

Suppose 1,000 probes are evaluated against a fixed 100-speaker gallery. The correct speaker is rank 1 for 920 probes and within the top 5 for 975 probes. Then

$$\text{Top1} = 920/1000 = 92.0\%, \quad (35)$$

$$\text{Top5} = 975/1000 = 97.5\%. \quad (36)$$

If the weakest 10 speakers have only 70% recall while the largest classes dominate the sample count, macro recall will reveal the problem even if total accuracy remains high.

12.2 Open-set example

Let $|K| = 3000$ known probes and $|U| = 3000$ unknown probes. At threshold θ :

- 2700 known probes are both correctly Top-1 matched and above threshold;
- 3 unknown probes exceed threshold for some gallery speaker.

Then

$$\text{DIR}(\theta, 1) = 2700/3000 = 0.900, \quad (37)$$

$$\text{FAR}(\theta) = 3/3000 = 0.001. \quad (38)$$

The system therefore achieves

$$\text{DIR@FAR} = 10^{-3} = 90.0\%. \quad (39)$$

This statement is incomplete unless gallery size and number of enrollment utterances are also reported.

12.3 Verification-cost example

Let $P_{tar} = 0.01$, $C_{miss} = 1$, $C_{fa} = 10$, and at a threshold let $P_{miss} = 0.05$, $P_{fa} = 0.001$. Then

$$C_{det} = 1(0.05)(0.01) + 10(0.001)(0.99) \quad (40)$$

$$= 0.0005 + 0.0099 = 0.0104. \quad (41)$$

The example shows how a small false-alarm probability can dominate cost when false accepts are expensive and non-target trials are common.

13 Statistical uncertainty and significance

13.1 Confidence interval for accuracy

If probe decisions were independent Bernoulli trials, standard error for observed accuracy $\hat{p}$ would be

$$SE \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{N}}. \quad (42)$$

A normal approximation gives $\hat{p} \pm 1.96SE$, but Wilson or exact intervals are preferable near 0 or 1 or for small samples.

Speaker trials are often correlated because multiple utterances come from the same speaker/session. Therefore **speaker-level or session-level bootstrap confidence intervals** are usually more defensible than treating every pair or segment as independent.

13.2 Paired system comparison

Two models evaluated on the same closed-set probes should be compared with a paired test. McNemar’s test uses only probes on which the two systems disagree. For ranking/DIR/EER, paired bootstrap resampling by speaker is often straightforward and preserves correspondence.

13.3 The NIST “Rule of 30” heuristic

Historical NIST speaker-recognition documentation cites Doddington’s “Rule of 30”: to be about 90% confident that the true error rate is within roughly $\pm 30\%$ of the observed error rate, one should observe at least 30 errors [12]. It is a heuristic, not a replacement for modern confidence intervals, but it conveys an important warning: an impressive low error rate estimated from only a few observed errors is statistically fragile.

14 Condition-stratified performance indicators

A single aggregate number should be accompanied by condition curves or tables.

14.1 Probe duration

Report Top-1 or DIR@FAR in duration bins, for example

$$[0.5, 1), [1, 2), [2, 4), [4, 8), \geq 8 \text{ s.} \quad (43)$$

Short-duration degradation often reveals a system that relies on long-term statistics.

14.2 SNR and acoustic environment

Report performance against SNR, reverberation time, or real environment category. Do not rely only on synthetic noise augmentation tests.

14.3 Channel/device

Separate matched-device and cross-device enrollment/probe conditions. NIST evaluations emphasize domain and channel mismatch because it strongly affects speaker recognition [7, 5].

14.4 Language and cross-lingual condition

Text-independent systems can still encode phonetic/language cues. SRE21 explicitly includes cross-language conditions [7]. Report same-language versus cross-language performance when deployment is multilingual.

14.5 Enrollment count

Report 1, 3, and 5 enrollment utterances or another deployment-relevant range. The prototype variance reduction can be dramatic in open-set identification [2].

14.6 Gallery size

At minimum, evaluate small, expected, and stress-test gallery sizes. If the production gallery can grow over time, performance as a function of N_G is more useful than one fixed benchmark result.

15 Metric pitfalls and common reporting errors

Table 1: Common metric mistakes and corrective practice.

Mistake	Why it is misleading	Better practice
Report only Top-1	No unknown rejection or rank profile	Add CMC; for open set add DIR@FAR
Report only EER	EER is 1:1 verification and threshold-optimized	Add 1:N identification metric and gallery size
Compare DIR without FAR	A lower threshold can trivially raise DIR	Compare at the same fixed FAR
Compare models with different galleries	Search difficulty changes with N_G	Freeze gallery/probe protocol
Ignore enrollment count	More enrollment samples improve prototypes	Report M_E and aggregation method
Use test set to tune threshold	Optimistically biases final result	Calibrate on a disjoint development set
Use accuracy with severe class imbalance	Majority speakers dominate	Add macro recall/F1 and per-speaker results
Pool every trial as independent	Same-speaker/session trials are correlated	Bootstrap or stratify at speaker/session level
Compare paper EERs as architecture ranking	Training data, LMFT, score norm, fusion differ	Re-run under common protocol
Use t-SNE separation as evidence	Visualization distorts geometry	Use task metrics; embedding plots only for diagnosis
Use spoof EER alone for secure access	CM and identity system interact	Add tandem cost/end-to-end attack acceptance

16 Recommended metric sets by application

Table 2: Recommended minimum evaluation indicators.

Application	Primary metrics	Secondary/diagnostic metrics
Fixed closed-set roster	Top-1, Top-5/CMC, macro recall	confusion matrix, duration/SNR strata, CI
Dynamic enrolled roster, known-only	Top-1/CMC from embedding gallery	verification EER/minDCF, search latency
Open-set access control	DIR@FAR at operational FAR; gallery-size sweep	EER/minDCF, calibration, enrollment-count sweep
Large watchlist	DIR@FAR at very low FAR; FPIR/TPIR; gallery-size curve	tail score analysis, calibration/fusion, ANN recall
Forensic / risk-based scoring	calibrated likelihood ratios, C_{lr} , application DCF	EER/minDCF, Bayes error-rate plots
Anti-spoof access	bona-fide DIR@FAR + t-DCF	CM EER, attack-specific false accept
Edge device	task accuracy + latency + RAM/energy	model size, real-time factor, worst-case duration

Application	Primary metrics	Secondary/diagnostic metrics
Research embedding paper	EER/minDCF plus closed/open SID metric if claiming identification	ablations, CI, cross-domain results

17 A recommended evaluation report template

For a deep speaker-identification model, publish one table for each of the following layers.

17.1 Protocol table

- train/dev/test speakers and sessions;
- N_G , M_E , $|K|$, $|U|$;
- duration statistics;
- scoring and threshold policy;
- dataset overlap controls.

17.2 Primary accuracy table

For closed set:

$$\{\text{Top1}, \text{Top5}, F1_{macro}, \text{BAcc}\}. \quad (44)$$

For open set:

$$\{\text{DIR}@10^{-3}, \text{DIR}@10^{-2}, \text{DIR}@10^{-1}\}. \quad (45)$$

Also report the gallery size and enrollment count in the table header.

17.3 Embedding diagnostic table

$$\{\text{EER}, \text{minDCF}, \text{actual DCF}, C_{\text{llr}}\} \quad (46)$$

when scores are calibrated or expected to be used as likelihood ratios.

17.4 Robustness table

Break the primary metric down by:

- duration;
- SNR/environment;
- enrollment device versus probe device;
- language/domain;
- enrollment count;
- gallery size.

17.5 Uncertainty

Add 95% confidence intervals, preferably from speaker/session-level bootstrap where the trial structure is correlated.

18 Metric selection for deep-learning model development

During training, the metric used for early stopping need not be the final deployment metric, but it should be aligned with the goal.

- For a fixed closed-set classifier, validation Top-1 or macro recall is appropriate.
- For a reusable embedding encoder, validation EER/minDCF is efficient during training, but periodic identification evaluation should be included.
- For an open-set production system, the final model should be selected with a development DIR@FAR at the intended gallery size and enrollment count, not only by EER.
- If threshold transfer matters, include calibration metrics; otherwise a model may require a new threshold for every domain.
- If spoof resistance matters, model selection should consider tandem system risk rather than identity accuracy alone.

19 Interpretation of benchmark results

19.1 VoxCeleb closed-set identification

The original VoxCeleb work used Top-1 and Top-5 accuracy for 1,251-class identification and reported that its CNN substantially outperformed traditional baselines [1]. The metric is appropriate because the task is explicitly closed-set. It does not demonstrate unknown-speaker rejection.

19.2 VoxBlink2 open-set identification

VoxBlink2 defines galleries of 60, 600, and 6,000 speakers, known and unknown queries, and 1/3/5 enrollment utterances per speaker. Its evaluation computes cosine similarities, requires the known speaker to be the correct Top-1 match, sweeps a threshold on the maximum unknown score, and reports DIR at specified FARs [2, 3]. The published results show that more enrollment utterances improve DIR, while a 6,000-speaker gallery can be dramatically harder at $\text{FAR} = 10^{-3}$ than a 60-speaker gallery.

19.3 VoxWatch

VoxWatch focuses on open-set watchlist detection and emphasizes that growing watchlists increase maximum out-of-set scores, creating a false-alarm problem; it also finds that calibration and score fusion can materially help, whereas adaptive score normalization is not guaranteed to help in that setting [4]. This supports evaluating calibration and watchlist size jointly rather than assuming verification post-processing transfers unchanged to identification.

20 Final recommendations

For most deep-learning speaker-identification research, the minimum defensible reporting set is:

1. **Closed-set:** Top-1 + Top-5/CMC + macro recall/F1 + confidence interval.
2. **Open-set:** DIR@FAR at at least two operational FAR values, with gallery size and enrollment count explicitly stated.
3. **Embedding diagnostic:** EER + minDCF + DET/ROC plot under a standard verification trial list.
4. **Calibration, if thresholds must transfer:** actual DCF and C_{llr} (or an equivalent proper scoring/calibration evaluation).
5. **Security, if identity decisions gate access:** spoofing metrics and preferably t-DCF/end-to-end attack acceptance.
6. **Robustness:** duration, SNR, device/domain, language, and gallery-size stratification.
7. **Statistics:** confidence intervals and paired significance testing for model comparisons.

The most important conceptual rule is:

$$\boxed{\text{verification quality} \neq \text{closed-set identification accuracy} \neq \text{open-set identification performance.}} \quad (47)$$

A modern speaker embedding can support all three, but each task requires its own evaluation protocol and primary indicator.

21 Conclusion

Speaker-identification accuracy is inherently multidimensional. Top-1 accuracy is the natural headline measure for a closed set, but CMC and macro class metrics reveal rank and speaker-specific behavior. In an open set, the system must jointly identify and reject; DIR@FAR (or equivalent TPIR@FPIR) is therefore the primary operational measure. EER, DET curves, and minDCF remain important because they measure the 1:1 discrimination of the underlying speaker embedding, while C_{llr} and actual DCF diagnose calibration. t-DCF extends evaluation to spoof-aware access systems. Finally, no metric should be interpreted without the gallery size, number of enrollment utterances, probe duration, threshold-selection method, and statistical uncertainty. For deep-learning speaker identification, a complementary metric suite is therefore not unnecessary complexity; it is required to distinguish genuine generalization from a result that is specific to one roster, one threshold, or one laboratory condition.

A Compact formula reference

$$\begin{aligned}
\text{Top1} &= \frac{1}{N} \sum_i \mathbb{I}(r_i = 1), \\
\text{Top}k &= \frac{1}{N} \sum_i \mathbb{I}(r_i \leq k), \\
\text{CMC}(k) &= \frac{1}{N} \sum_i \mathbb{I}(r_i \leq k), \\
\text{DIR}(\theta, k) &= \frac{|\{p \in K : r(p) \leq k, s(p, g_p) \geq \theta\}|}{|K|}, \\
\text{FAR}(\theta) &= \frac{|\{u \in U : s^*(u) \geq \theta\}|}{|U|}, \\
P_{fa}(\theta) &= P(s \geq \theta \mid \text{non}), \\
P_{miss}(\theta) &= P(s < \theta \mid \text{tar}), \\
C_{det}(\theta) &= C_{miss}P_{miss}P_{tar} + C_{fa}P_{fa}(1 - P_{tar}), \\
\text{minDCF} &= \min_{\theta} C_{norm}(\theta).
\end{aligned}$$

B Checklist for an evaluation section in a paper

1. State closed-set or open-set explicitly.
2. State gallery size and enrollment count.
3. State train/dev/test speaker overlap policy.
4. State probe duration and VAD rules.
5. State scoring method and score normalization.
6. State threshold-selection set and target FAR.
7. Report Top-1/CMC for closed set.
8. Report DIR@FAR for open set.
9. Report EER/minDCF only as supporting verification diagnostics unless the task is verification.
10. Add calibration metric when a fixed threshold or probability is operationally important.
11. Add attack-aware metric for secure access.
12. Add confidence intervals and condition-stratified results.

References

- [1] A. Nagrani, J. S. Chung, and A. Zisserman, “VoxCeleb: A Large-Scale Speaker Identification Dataset,” *Proc. Interspeech*, 2017.
- [2] Y. Lin, M. Cheng, F. Zhang, Y. Gao, S. Zhang, and M. Li, “VoxBlink2: A 100K+ Speaker Recognition Corpus and the Open-Set Speaker-Identification Benchmark,” arXiv:2407.11510, 2024.
- [3] VoxBlink2 authors, “ScriptsForVoxBlink2: official OSSI evaluation implementation,” GitHub repository, 2024–2026.
- [4] R. Peri, S. O. Sadjadi, and D. Garcia-Romero, “VoxWatch: An Open-Set Speaker Recognition Benchmark on VoxCeleb,” arXiv:2307.00169, 2023.
- [5] G. R. Doddington, M. A. Przybocki, A. F. Martin, and D. A. Reynolds, “NIST Speaker Recognition Evaluation – Overview, Methodology, Systems, Results, Perspective,” *Speech Communication*, 2000.
- [6] A. F. Martin and M. A. Przybocki, “NIST Speaker Recognition Evaluation Chronicles,” NIST technical publication.
- [7] S. O. Sadjadi, C. Greenberg, E. Singer, L. Mason, and D. Reynolds, “NIST 2021 Speaker Recognition Evaluation Plan,” National Institute of Standards and Technology, 2021.
- [8] N. Brümmer and E. de Villiers, “The BOSARIS Toolkit: Theory, Algorithms and Code for Surviving the New DCF,” arXiv:1304.2865, 2013.
- [9] N. Brümmer and J. du Preez, “Application-Independent Evaluation of Speaker Detection,” *Computer Speech & Language*, vol. 20, no. 2–3, pp. 230–275, 2006.
- [10] T. Kinnunen, K. A. Lee, H. Delgado, N. Evans, M. Todisco, M. Sahidullah, J. Yamagishi, and D. A. Reynolds, “t-DCF: a Detection Cost Function for the Tandem Assessment of Spoofing Countermeasures and Automatic Speaker Verification,” *Proc. Odyssey*, 2018.
- [11] S. Seo, O. Aulov, A. Godil, and K. Mangold, “Evaluating Identity Leakage in Speaker De-Identification Systems,” NIST / ICASSP 2026 publication, 2025.
- [12] NIST, “The NIST Speaker Recognition Evaluations: 1996–2001,” technical retrospective including Doddington’s Rule of 30.